

25. NOT FOR SUBMISSION

Practice computing determinants. For example, choose some “sufficiently random” matrices $A \in \mathbb{R}^{3 \times 3}$ and $B \in \mathbb{R}^{4 \times 4}$, and compute their determinants using standard methods and propositions (explicit rules for low-order matrices, Laplace expansion, elementary row/column operations, reduction to triangular form, etc.). Aim to choose an efficient combination of methods in order to avoid tedious calculations. Verify your results using a computer algebra system / an online matrix calculator / AI tools / etc.

26. Compute the determinants of the following matrices. Use relevant propositions and properties of determinants to carry out all calculations efficiently and easily by hand, without using a calculator / computer algebra system / etc.

$$A = \begin{pmatrix} 234 & 235 & 236 \\ 235 & 236 & 237 \\ 236 & 237 & 239 \end{pmatrix}, \quad B = \begin{pmatrix} 21 & 23 & 0 & 27 & 2 \\ 0 & 2 & 0 & 7 & 0 \\ 11 & 12 & 13 & 14 & 15 \\ 1 & 41 & 0 & 43 & 0 \\ 0 & 1 & 0 & 4 & 0 \end{pmatrix}.$$

27. Let

$$A = \begin{pmatrix} a & 1 & 2 & 3 \\ 1 & 2 & 3 & 2 \\ 2 & 3 & 2 & 1 \\ 3 & 2 & 1 & b \end{pmatrix} \in \mathbb{R}^{4 \times 4},$$

where a and b are real parameters.

- (a) Compute the determinant of A .
 - (b) Find the condition on a and b under which the system $A\mathbf{x} = 0$ has infinitely many solutions.
28. Let $n \geq 3$ be an odd number, and let $A \in \mathbb{R}^{n \times n}$ be a skew-symmetric matrix. Prove that the rows of A are linearly dependent.
29. Let $n \geq 3$, and let $A \in \mathbb{R}^{n \times n}$ be a matrix in which every row forms an arithmetic sequence. Prove that $\det(A) = 0$. Try to give several arguments (possible keywords: linearly dependent rows, nontrivial solution of $A\mathbf{x} = 0$, the dimension of the row space, ...).
30. Consider the following system of linear equations:

$$\begin{cases} 2x_1 + 5x_2 + 3x_3 = 5 \\ -x_1 - 4x_2 + x_3 = -3 \\ 2x_2 - 5x_3 = -1 \end{cases}$$

Solve this system in two ways:

- (a) Using the adjugate matrix,
- (b) Using Cramer’s rule.